COURSE MATERIAL	
SUBJECT	ENGINEERING GRAPHICS (ME23AES102)
UNIT	5
COURSE	B.TECH
SEMESTER	11
DEPARTMENT	MECHANICAL ENGINEERING
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1. COURSE OBJECTIVES

The objectives of this course is to introduce and make the students

- To enable with various concepts like dimensioning, conventions and standards related to Engineering Drawing
- To impart knowledge on the Projection of Points, Lines and Plane surfaces
- To improve the visualization skills for better understanding of projection of solids
- To develop the imaginative skills required to understand Section of solids and Developments of surfaces.
- To understand the viewing perception of a solid object in Isometric and orthographic projections.

2. PREREQUISITES

Students should have knowledge on

Basic Mathematics

3. SYLLABUS

UNIT V

Conversion of Views: Conversion of Isometric Views to Orthographic Views of Simple Solids; Conversion of Orthographic Views to Isometric views of Simple Solids.

4. COURSE OUTCOMES

After completing the course, the student will be able to

1. Draw various engineering curves, scales. (L3)
2. Draw and Interpret orthographic projections of points, lines, planes. (L3)
3. Draw the projection of solids in various positions. (L3)
4. Draw and Explore the sections of solids and development of surfaces. (L3)
5. Draw an isometric and orthographic views of simple solids. (L3).

5. Co-PO / PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2								2		2	2	2
CO2	2	2								2		2	2	2
CO3	2	2								2		2	2	2
CO4	2	2								2		2	2	2
CO5	2	2			1					2		2	2	2

6. LESSON PLAN

LECTURE	WEEK	TOPICS TO BE COVERED	REFERENCES
1	1	Isometric Projection	T1
2		Lab practice	T1, R1
3	2	Isometric Projection	T1, R1
4		Lab practice	T2, R1
5	3	Orthographic Projection	T1, R1
6		Lab practice	T2, R1

7. LECTURE NOTES

5.1 Orthographic Projections

In orthographic projection, an object is represented by projecting its views on imaginary orthogonal planes. Any object, irrespective to the dimensions, (1D, 2D or 3D objects) is converted to 2D drawings or projections.

5.1.1 Reference planes

Principal planes - horizontal plane and vertical plane –are the main reference planes used in orthographic projections.

Profile plane, auxiliary vertical plane, and auxiliary inclined plane are also used as reference planes when two views of the object are not sufficient.

5.1.2 Principal Planes

Horizontal and Vertical planes are the principal planes used in orthographic projections (refer Figure 5.1).

5.1.3 Horizontal Plane (HP)

A plane of reference which is assumed to be parallel to the plane along the horizon or a plane which is perpendicular to the gravity field at a place. In orthographic projection, there is only one horizontal plane.

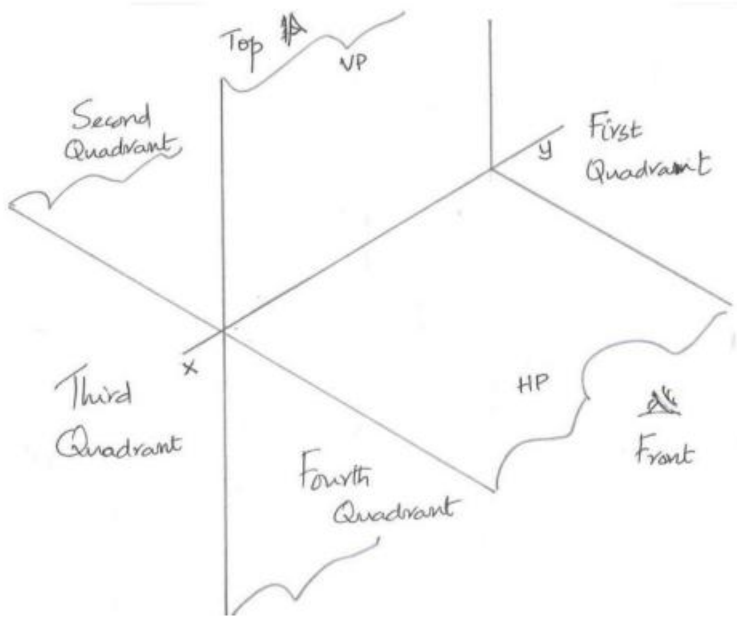


Figure 5.1 Principal Planes and Quadrants

5.1.4 Vertical Plane (VP)

A reference plane which is assumed to be along or parallel to the gravity field. This plane will be perpendicular to the horizontal plane.

5.1.5 Auxiliary Planes

Auxiliary Vertical Plane (AVP)

Planes perpendicular to HP but inclined to VP are AVPs. Projection on an AVP is called as auxiliary front view

5.1.6 Main Reference Line, x-y

The line of intersection of Horizontal plane (HP) and Vertical plane (VP) is the main reference line, x-y (refer Figure 5.1).

The lines of intersection of auxiliary planes with principal planes are auxiliary reference lines. While drawing auxiliary projections, auxiliary reference lines are used for representing projections

Basic characteristics of planes of projection

- Planes are assumed to have enormous area so that any object irrespective to the size can be projected to the plane.
- Planes do have negligible thickness, such that they appear as lines while observing along the plane.
- Planes are assumed to be transparent, so that irrespective to the quadrant where the object situates the observer can view it directly from both front and top.
- Planes are not rigidly attached to the other plane(s), such that one plane can be rotated about the line of intersection to make coinciding with the other.

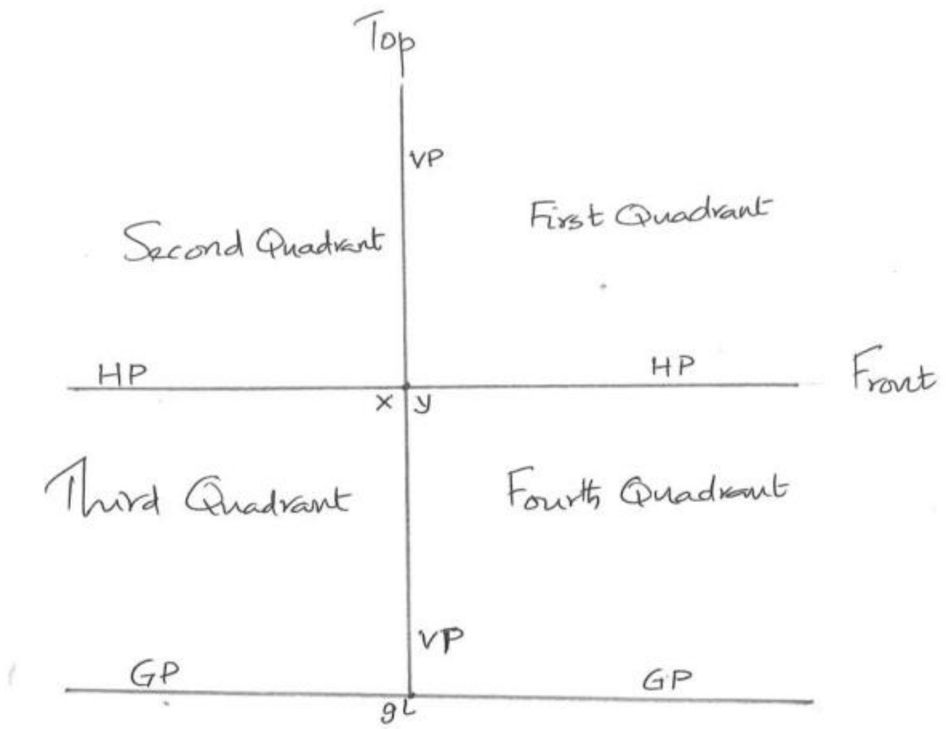


Figure 5.2 Principal Planes, Ground Plane and Quadrants viewing along x-y

5.3 First Angle Projection

Projection of any object drawn assuming the object in the first quadrant is First Angle Projection. As per recommendation of Bureau of Indian Standards First angle projection is followed in India.

5.4 Third Angle Projection

Projection of any object drawn assuming the object in the third quadrant is Third Angle Projection.

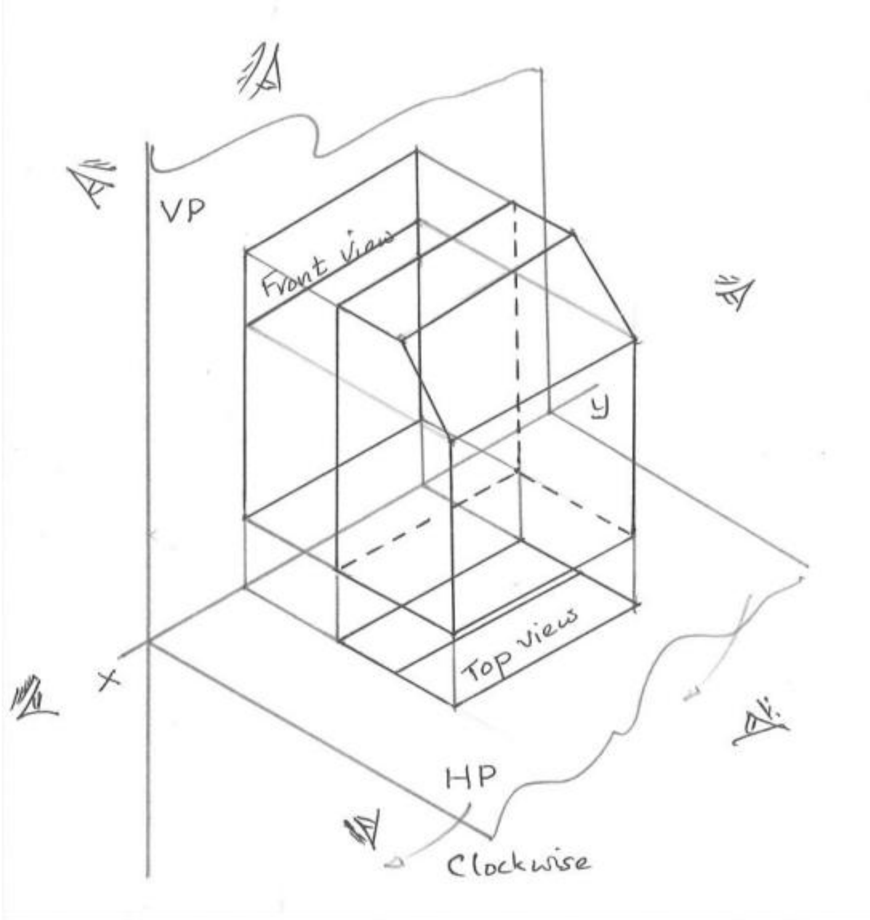


Figure 5.3 Directions of Observation of an Object in the First Quadrant

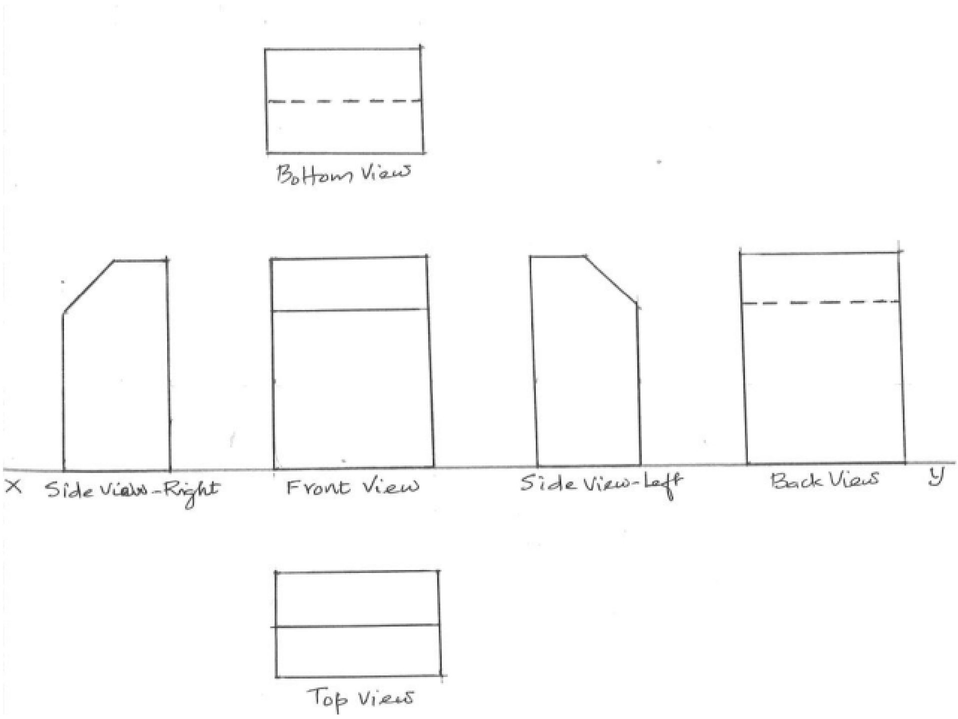
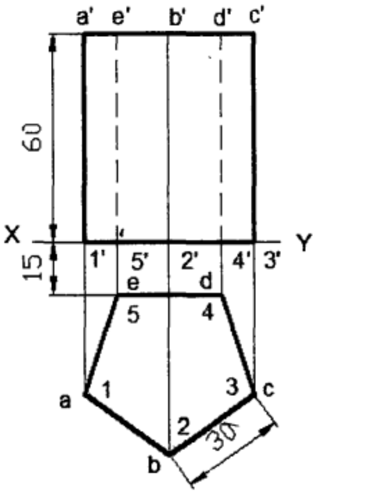
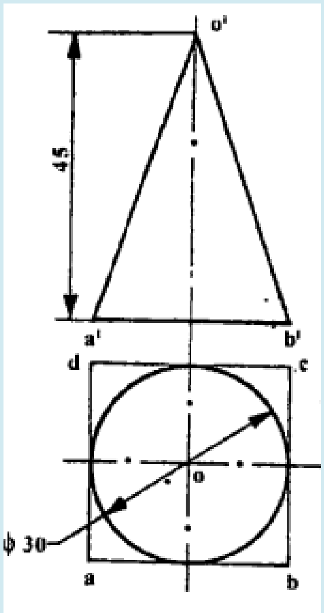


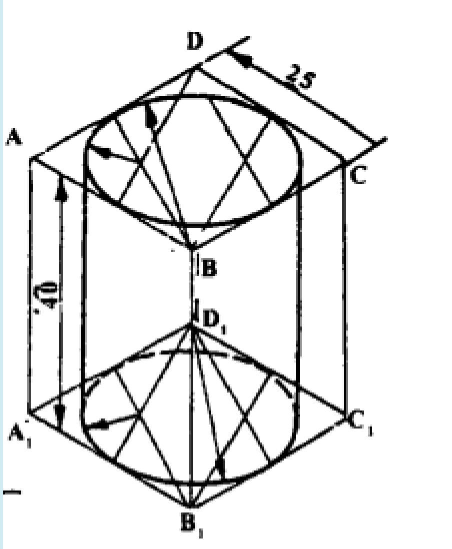
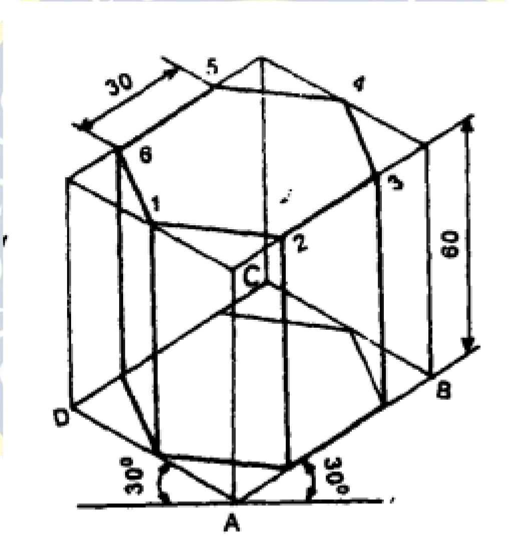
Figure 5.4 Six Views of the Object in Figure 5.3

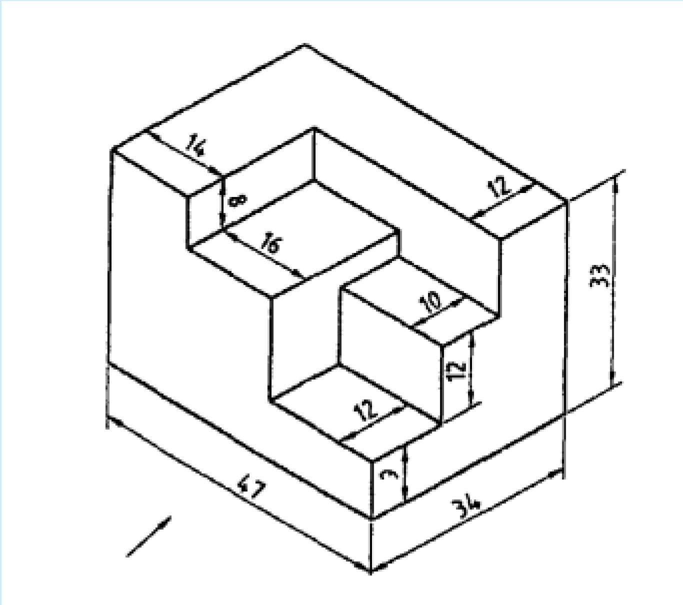
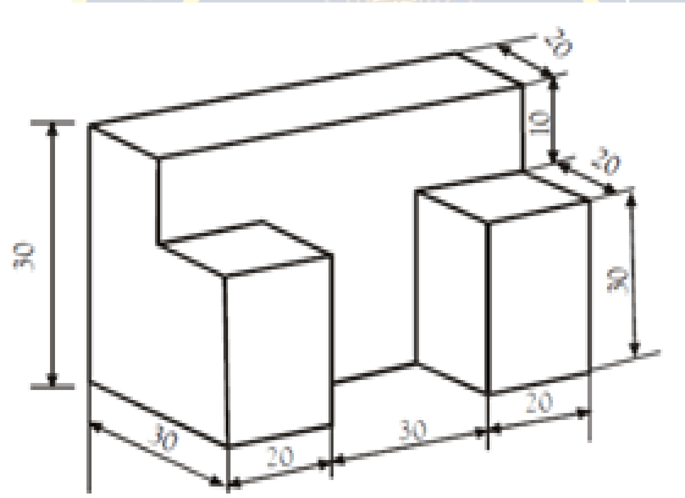
8. Assignments

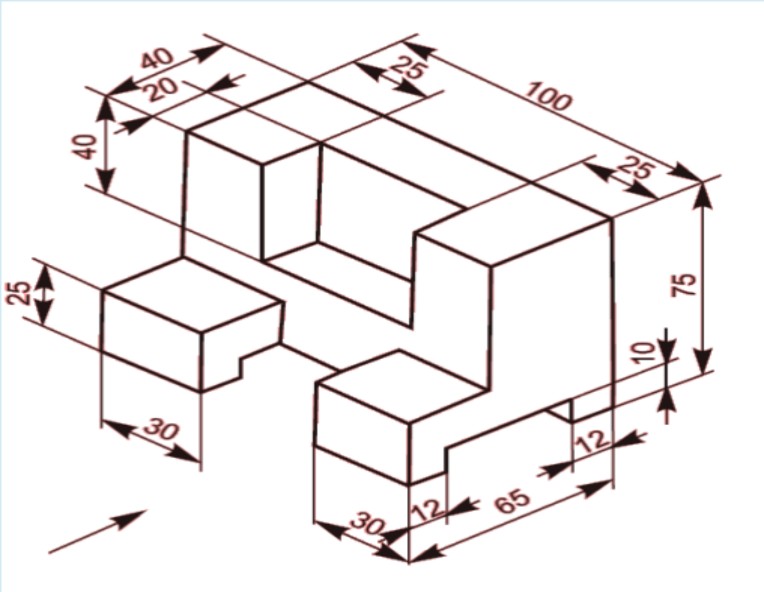
S.No	Question	BL	CO
1	Draw the isometric view of a square prism, side of base 30mm long and axis 55mm long when its axis is a) Vertical b) Horizontal	1	5
2	A hexagonal prism with a 30 mm base and 45 mm axis. Draw its isometric view	1	5
3	Draw the isometric view of a triangular pyramid of side 30 mm and axis height 60 mm.	1	5
4	Draw the isometric view of a the following figure. 	1	5
5	Draw the isometric view of a the following figure 	1	5

9. Part B- Questions

S.No	Question	BL	CO

1	Draw the front view and top view of the drawing given in figure below. All the dimensions in mm  	1	5
2	Draw the orthographic views of the solid shown in figure below  	1	5

3	Draw the front view, top view and side view of the drawing given in figure below. All the dimensions in mm  	1	5
4	Draw the front view, top view and side view of the drawing given in figure below. All the dimensions in mm  	1	5
5	Draw the front view, top view and the right side view of the drawing given in figure below. All the dimensions in mm	1	5



10. Supportive Online Certification Courses

1. Engineering drawing and computer graphics By Prof. Rajaram Lakkaraju, conducted by IIT Kharagpur – 12 weeks

11. Prescribed Textbooks & Reference Books

Textbook

1. K.L. Narayana & P. Kannaiah, Engineering Drawing, 3/e, SciTech Publishers, Chennai, 2012.
2. N.D. Bhatt, Engineering Drawing, 53/e, Charotar Publishers, 2016.

References:

1. Venugopal, Engineering Drawing and Graphics, 3/e, New Age Publishers, 2000.
2. K.C. John, Engineering Graphics, 2/e, PHI, 2013.